MIL-V-4719B(USAF)
16 July 1984
SUPERSEDING
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21 Apr 1967

MILITARY SPECIFICATION

VALVES, MANUALLY OPERATED, GAS SERVICING HOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of valves for gas servicing hose.

1.2 Classification. Valves shall be of the following three types as specified (see 6.2).

Type I - Low Pressure Oxygen, servicing up to 2200 pounds per square inch gage (psig).

Type II - High Pressure Oxygen, servicing 2200 to 4000 psig.

Type III - Air on Nitrogen, servicing up to 4000 psig.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-T-634	Trichloroethylene, Technical
P-C-436	Cleaning Compound, Alkali, Boiling Vat (Soak) or Hydrosteam.
BB-N-411	Nitrogen, Technical.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3655

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QQ-P-416	Plating, Cadmium (Electrodeposited).
QQ-B-637	Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bars, Flat Wire, and Strip).
QQ-S-763	Steel, Corrosion-Resisting; Bars and Forgings (Except for Reforging).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-665	Boxes, Paperboard, Metal Edged and Components.
PPP-B-676	Boxes, Setup.

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DOD-D-1000	Drawing, Engineering and Associated Lists.
MIL-P-116	Preservation, Methods of.
MIL-B-117	Bag, Sleeve and Tubing, Interior Packaging.
MIL-B-131	Barrier Material, Watervaporproof, Flexible, Heat-Sealable.
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for.
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for.
MIL-C-15074	Corrosion Preventive, Fingerprint Remover.
MIL-C-81302	Cleaning Compound, Solvent, Trichlorotifluoroethane.

STANDARDS

FEDERAL

FED-STD-101	Test Procedures for Packaging Materials.

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of.
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MS20913	Plug, Square Head Pipe Thread.
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal.

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions shojdl be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents for a part of this specification to the extent specified herein.

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DRAWINGS

AIR FORCE

52B3962 Gun Assembly-Ignition, Material Testing with Gases.

AIR FORCE-NAVY AERONAUTICAL

AND10089 Fitting End, Standard Dimension for Cone Connection.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 3951 Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) first article inspection shall be performed as specified in 4.3.

3.2 Specification and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Nonferrous materials. Nonferrous materials shall be used for all parts of the valve, except where required by this document and where ferrous materials are necessary.

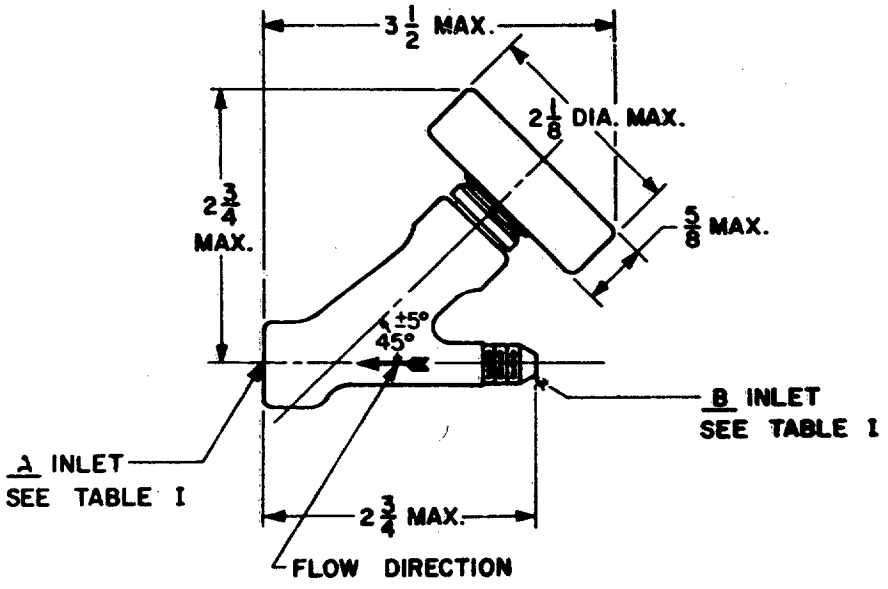
3.3.2 Nonmetallic materials. Any nonmetallic materials that is easily deteriorated or otherwise affected adversely by any of the constituents of air shall not be used in the construction of the valve.

* 3.3.3 Valve body. The valve body shall be made of forged naval brass in accordance with QQ-B-637 or stainless steel, class 304L in accordance with QQ-S-763 or equal.

3.3.4 Valve seat disc. The valve seat disc material shall be resistant to ignition by sudden compression of the pressurized gas.

3.3.5 Handwheel. The handwheel shall have a metal core with a bonded rubber gripping surface in accordance with Figure 1. The handwheel shall be marked to show the direction of opening.

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NOTES:
1. Valve body or at least one end fitting must have wrenching surface.
2. Dimensions in inches.

FIGURE 1. Valves, manually operated gas servicing hose.

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3.3.6 Plugs. All openings in the valve shall be closed with caps or plugs to prevent dust and other foreign matter form entering the valve during shipment and storage. The threaded plugs shall conform to MS20913, except that the material shall be plastic or soft synthetic rubber. The material and design of all other caps or plugs shall be subject to the approval of the procuring agency.

3.4 Design and construction. The valve shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Valve construction. The valve shall be constructed so that adjustments and repairs can be easily made by the personnel of operating units.

3.4.2 Valve design. The valve shall be designed so that it is fully opened with not more than 2 counterclockwise (ccw) turns of the handwheel. The valve shall be of the packless, positive shutoff type so that the stem is activiated through the use of suitable metallic diaphragms; or of the packed, positive opening and closing type utilizing a needle-point seated stem. Metallic bellows shall not be used. Wrench gripping surfaces not smaller than 3/8 inch wide and 3/4 inch in the direction of gas flow shall be provided. When the valve is of the packed type, the packing utilized shall be suitable for use with any of the constituents of air under high pressure. The valve shall be designed to close completely with a torque not exceeding 60 pound-inches when of the diaphragm type and 100 pound-inches when of the packed type.

3.4.3 Operating pressure. The valve shall be designed for a maximum operating pressure as outlined in Table I. It shall be capable of withstanding a hydrostatic pressure test of not less than 1 1/2 times maximum working pressure without deformation or leakage.

3.4.4 Gas channels. The "IN" part of the valve shall be connected to the channel which confines the gas under the valve seat while the valve is in the closed position.

3.4.5 Threads.

3.4.5.1 Screw. Straight screw threads shall be in accordance with MIL-S-7742.

3.4.5.2 Pipe. Pipe threads shall be in accordance with MIL-P-7105.

3.5 Performance.

3.5.1 Flow. The valve shall be capable of delivering 25 cubic feet per minute (cfm) of the free gas at 100 psig inlet pressure when fully opened and with atmospheric pressure at the outlet. (see 4.4.2.4)

3.5.2 Endurance. The valve shall be capable of withstanding 1000 opening and closing cycles at a maximum specified operating pressure. (see 4.3.3.4)

3.5.3 Temperature. The valve shall be capable of operating in a temperature range of -65° to $+160^{\circ}$ Fahrenheit (F).

3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

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3.7 Dimensions. The dimensions of the valve shall conform to Figure 1 and Table I.

3.8 Weight. The weight of the valve shall not exceed one pound.

3.9 Ferrous materials. Ferrous materials, not including stainless steel, shall be cadmium plated in accordance with QQ-P-416, type II, class B.

TABLE I

Type	Outlet	Inlet	Maximum Operating Pressure
	A	B	
I	1/4 inch internal pipe thread	#5 size MS-33656	2200 psig
II	#3 size AND-10089	#5 size MS-33656	4000 psig
III	#4 size MS-33656	#4 size MS-33656	4000 psig

3.10 Degreasing. All integral parts of the valve shall be degreased by a trichlorethylene process or equal.

3.11 Marking.

3.11.1 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The identification plate shall be circular and shall be securely attached to the valve handle. The identification data shall be as follows:

Valve Type*
Specification MIL-V-4719B
Mfr's name or trademark

*Complete information to be withdrawn from 1.2.

3.11.2 Direction marking. A circular arrow, preceded by the word "OPEN," showing the direction of opening, shall be inscribed on the nameplate.

3.12 Workmanship. The valve shall be constructed in a thoroughly workmanlike manner. The finished valve shall be clean and free from any defects that might affect appearance or functionability. The valves shall be thoroughly cleaned of all oil, grease, dirt, and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection.

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. Packaging inspection (see 4.7)

4.3 First article inspection. Four sample valves shall be furnished and subjected to the examination of 4.5 and the tests of 4.6. Any sample which fails any examination or test shall be cause for failure of first article inspection.

4.3.1 First article test report. If the tests are conducted by activities other than the procuring activity, the submitting firm shall be required to submit to the procuring activity three (3) complete copies of the Engineering Test Report prepared in accordance with MIL-STD-831.

4.4 Quality conformance inspection.

4.4.1 Lot. An inspection lot shall consist of all valves of one type offered for delivery at one time.

4.4.2 Sampling for quality conformance inspection. (Specify levels, AQL in accordance with MIL-STD-105, and state rejection criteria.

4.4.2.1 Sampling for examination.

4.4.2.2 First article test samples. The first article test samples shall consist of four valves representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.5 Examination. Each valve shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, design and marking.

4.6 Tests.

4.6.1 Test conditions.

4.6.1.1 Standard atmospheric conditions. Wherever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (29.92 inches mercury (hg)) and at room temperature (+70°F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.6.1.2 Test gas. The gas used in conducting the tests specified herein, unless otherwise specified, shall be nitrogen (water-pumped) conforming to BB-N-411 or water-pumped air equivalent in dryness to nitrogen.

4.6.2 Test procedures.

4.6.2.1 Seat material test. If other than metal seats are provided, seat material shall be tested in accordance with 52B3962 (test for ignition resistance of elastomers).

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4.6.2.2 High temperature test. The valve, while in the closed position and the outlet uncapped, shall be subjected to a high temperature test in accordance with Procedure I of MIL-STD-810. At the conclusion of the exposure period, the valve shall be subjected to maximum operating pressure, and tested for leakage and for ease of operation. The same test shall be conducted with the outlet of the valve capped and the valve in the open position.

4.6.2.3 Low temperature tests. The valve shall be subjected to a low temperature test in accordance with Procedure I of MIL-STD-810. The operation tests shall be conducted at maximum operating pressure at -65°F with the valve closed and outlet uncapped, then repeated with the valve opened and the outlet capped. There shall be no evidence of leakage during or at the completion of this testing.

4.6.2.4 Endurance tests. The valve shall be subjected to 1000 opening and closing cycles as follows: The outlet shall be capped and the valve subjected to a pressure of 1 1/2 times maximum operating pressure. The valve shall then be cycled at the rate of not less than one cycle per minute (cpm). During this test, the torque used to close the valve shall not exceed 60 pound-inches or 100 pound-inches, as applicable. Upon conclusion of the 1000 cycles, the valve shall be subjected to the test specified in 4.4.2.2.

4.6.2.5 Leakage test. The valve, while at normal temperature, shall be subjected to maximum operating pressure. The valve shall be turned to the open position with the outlet capped and then to the closed position with the outlet uncapped. No more than 60 pound-inches shall be required to close the valve if of the diaphragm type and 100 pound-inches if of the packed type. There shall be no evidence of leakage from any part of the valve when examined while immersed in water.

4.6.2.6 Flow test. With an inlet pressure of 100 psig and atmospheric pressure at the outlet, the valve, while in the fully opened position, shall permit a flow of 25 cfm at atmospheric pressure and normal temperature.

4.6.2.7 Hydrostatic test. The inlet of the valve shall be subjected to a water pressure of 1 1/2 times maximum operating pressure with the outlet capped and the valve in the opened position; then in the closed position with the outlet uncapped. Any evidence of deformation or leakage shall be cause for rejection.

4.6.2.8 Stress corrosion test. Valves using a body material of brass shall be subjected to a stress corrosion test. Two additional valves from the lot of 100 or fraction thereof, specified in 4.4.2, shall withstand immersion for 15 minutes without cracking in an aqueous solution containing 100 grams of mercurous nitrate and 13 milliliters (ml) of nitric acid (specific gravity 1.42) per liter. The valve shall be examined for cracks immediately upon removal. Any cracks revealed by either valve shall be cause for rejection of the lot.

4.7 Packaging inspection.

4.7.1 Unit of product. For the purpose of examination, a complete pack, prepared for shipment, shall be considered a unit of product.

4.7.2 Sampling for packaging inspection.

4.7.2.1 Examination of packaging. One sample pack (see 4.7) (state exam required and show classification of defects in Table I).

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4.7.3 Pack tests. One sample pack (see 4.7.1) shall be selected at random from each lot and subjected to the following tests. Failure of any test shall result in rejection of the lot.

a. Free fall drop test. The pack shall be drop tested in accordance with FED-STD-101, method 5007, procedure B and procedure E.

b. Cornerwise drop test. The pack shall be tested as specified in FED-STD-101, method 5005.

c. Edgewise drop test. The pack shall be tested in accordance with FED-STD-101, method 5008.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-P-116.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The valves shall be thoroughly cleaned and dried by one or a combination of the following methods.

5.1.1.1.1 Method A. Vapor degreasing process followed by fingerprint removal. Use stabilized trichloroethylene conforming to O-T-634 a standard commercial vapor degreaser. The operaton of the degreaser shall be in accordance with the manufacturer's recommendations. Fingerprint remover shall conform to MIL-C-15074. Following application of the fingerprint remover, surfaces shall be thoroughly rinsed with clean trichloroethylene. All solvents shall be removed after the degreasing operation by purging with hot oil-free air or nitrogen.

5.1.1.1.2 Method B. Solvent degreasing by slushing methods. All surfaces requiring degreasing shall be thoroughly washed with solvent conforming to MIL-C-81302. Degreased surfaces shall be cleaned with fingerprint remover conforming to MIL-C-15074. Following application of the fingerprint remover, the surfaces shall be thoroughly rinsed with clean trichloroethylene. All solvents shall be removed after the degreasing operation by purging with hot filtered oil-free air or nitrogen. Precautions shall be taken to insure the chlorinated solvent does not contact plastics or other incompatible materials.

5.1.1.1.3 Method C. Alkali detergent degreasing. Components shall be flushed with hot alkaline cleaning compound conforming to P-C-436 until free from oil, grease and other combustible materials. Following the alkaline cleaning, all surfaces, internal and external, shall be thoroughly rinsed with clean hot water and dried by purging with hot, filtered oil-free air or nitrogen, or by baking at a temperature of 250° to 300°F, until all water is removed.

5.1.1.2 Unit protection. Immediately after drying, valve shall be packaged one each by Method IA-8 in accordance with MIL-P-116. A preservative compound shall not be used. Bags shall be in accordance with MIL-B-131, class 1, MIL-B-117, type I. The bagged valves shall be individually packaged in containers conforming to PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676.

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5.1.1.3 Intermediate packaging. Ten valves packaged as described in 5.1.1.2 shall be placed in a close-fitting intermediate container conforming to the requirements of PPP-B-636, type CF, class domestic. Closure shall be in accordance with the appendix of PPP-B-636.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

5.2.1 Level A. A quantity of valves preserved as specified in 5.1 shall be packed in a wood-cleated plywood shipping container conforming to PPP-B-601, overseas type, Grade A, Style optional. Insofar as practical, containers shall contain identical quantities and be of minimum cube and tare consistent with the protection required. Closure and strapping shall be in accordance with the appendix to PPP-B-601.

5.2.2 Level B. A quantity of valves preserved as specified in 5.1 shall be packed in a wood-cleated plywood shipping container conforming to PPP-B-601, domestic type, Grade B, Style optional, or a fiberboard shipping container conforming to PPP-B-636, Class Weather Resistant, Style optional. Insofar as practical, containers shall contain identical quantities and be of minimum cube and tare consistent with the protection required. Closure and strapping shall be in accordance with the appendix to the applicable container specification.

5.2.3 Commercial. Valves shall be packed in accordance with the requirements of ASTM D 3951.

5.3 Physical protection. Shipping and intermediate containers shall be cushioned in accordance with MIL-STD-1186.

5.4 Marking. In addition to any special marking specified in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. The line valves covered by this specification are intended for use as servicing valves on recharging units for aircraft systems.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title and date of this specification
b. Type of valve required (see 1.2)
c. When first article is required (see 3.1)
d. Level of packaging required (see 5.2)

TABLE II

Type	Part Number
I	MIL-V-4719-1
II	MIL-V-4719-2
III	MIL-V-4719-3

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* 6.3 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractor's are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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